MACHINE-LEARNING BASED CUSTOMER SATISFACTION PREDICTION

Video:
https://drive.google.com/file/d/1C7UrWPILmHnXMwvCeDLvL2SKJkYAG2E-/view?usp=drive_link







AGENDA

- Research Question and Task
- Data Processing and Exploration
- Model Exploration and Evaluation
- Discussion

RESEARCH QUESTION

- The research question is to predict customer's satisfaction after purchase using existing information regarding costumers and products, including demographic factors such as age, gender, income, region, and loyalty status, along with promotional activities and product categories.

TASK

- Binary-classification task that predict whether customer's satisfied or not based their information.

Reference:

https://hbr.org/2023/03/how-machine-learning-can-improve-the-customer-experience

DATA OVERVIEW

Data Source:

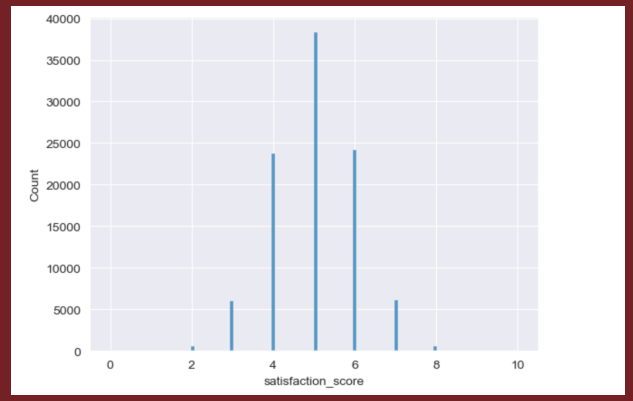
- Kaggle Dataset: https://www.kaggle.com/datasets/sanyamgoyal401/customer-purchases-behaviour-dataset
- This dataset contains simulated data representing customer purchase behavior. It includes various features such as age, gender, income, education, region, loyalty status, purchase frequency, purchase amount, product category, promotion usage, and satisfaction score.



DATA PROCESSING

➢ About the dataset:

- 10000 samples, 12 attributes




```
satisfaction_category
satisfied       30941
unsatisfied     30301
Name: count, dtype: int64
```

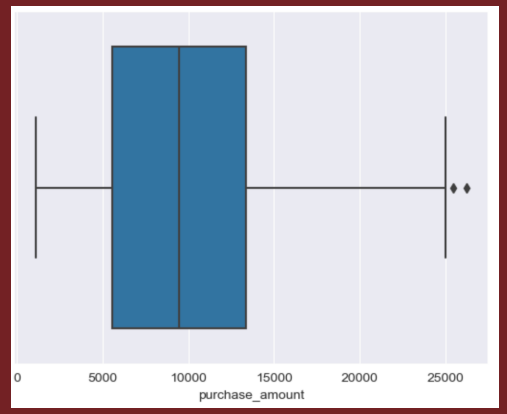
- Target variable: Satisfaction category
  - Created from satisfaction score: score>5 → satisfied, score<5 →unsatisfied, =5 → dropped

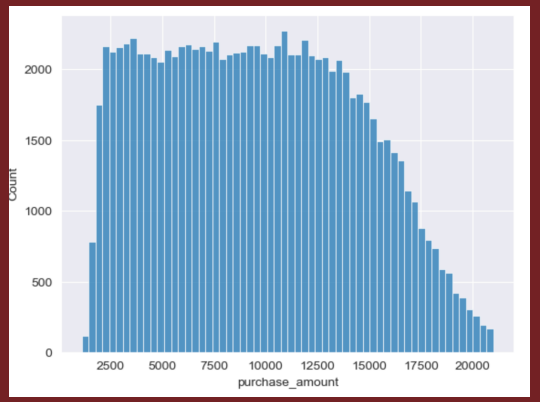
- Features:
  - Did not select id
  - Customer features: age, gender, education, region, loyalty_status, purchase_frequency, purchase_amount
  - Product features: product_categoty, promotion_usage

DATA CLEANING AND TRANSFORMATION

➢ Missing values and outliner detection:

- No missing values
- Few outliners
  - For ‘purchase amount’: dropped when >21000





➢ Normalize Numerical Data

- For ‘income’ and ‘purchase amount’
- normalization doesn't change the distribution

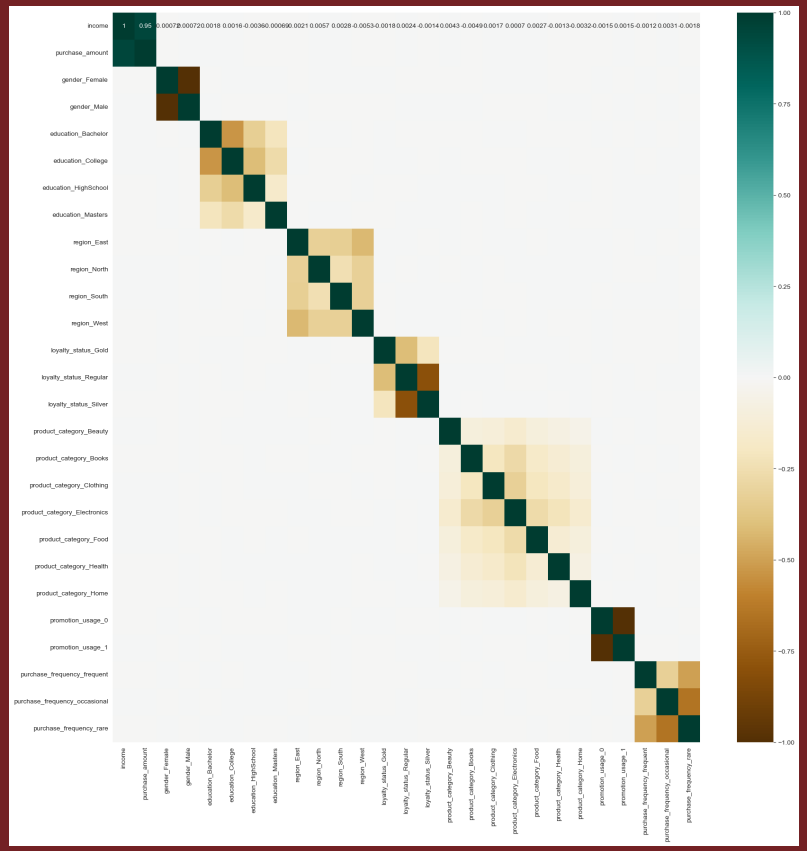
▪ Encode Categorical Data

- For: gender, education, region, loyalty_status, purchase_frequency, product_categoty, promotion_usage
- One-hot Encoder

EXPLORATORY ANALYSIS

Collinearity Check:

- We don’t observe obvious linear relationship among features.



MODEL EXPLORATION AND EVALUATION

- ✓ Supervised learning
- ✓ Unsupervised learning
- ✓ Ensemble techniques



DECISION TREE

1. Train, fit and evaluate the model
2. Hyper-parameter tuning
3. Result Analysis

Baseline performance with default hyperparameters

	precision	recall	f1-score	support
satisfied	0.50	0.58	0.54	9173
unsatisfied	0.51	0.42	0.46	9200
accuracy			0.50	18373
macro avg	0.50	0.50	0.50	18373
weighted avg	0.50	0.50	0.50	18373

Best-fit model performance with optimal hyperparameters

	precision	recall	f1-score	support
satisfied	0.50	0.77	0.61	9173
unsatisfied	0.51	0.24	0.33	9200
accuracy			0.50	18373
macro avg	0.51	0.50	0.47	18373
weighted avg	0.51	0.50	0.47	18373

❖ Hyper-parameter tuning helps improve the performance for satisfied group, but not for the unsatisfied group.



LOGISTIC REGRESSION

1. Train, fit and evaluate the model
2. Hyper-parameter tuning
3. Result Analysis

Baseline performance with default hyperparameters

	precision	recall	f1-score	support
satisfied	0.50	0.80	0.62	9173
unsatisfied	0.51	0.20	0.29	9200
accuracy			0.50	18373
macro avg	0.50	0.50	0.45	18373
weighted avg	0.50	0.50	0.45	18373

Best-fit model performance with optimal hyperparameters

	precision	recall	f1-score	support
satisfied	0.50	0.84	0.63	9173
unsatisfied	0.52	0.17	0.25	9200
accuracy			0.50	18373
macro avg	0.51	0.51	0.44	18373
weighted avg	0.51	0.50	0.44	18373

- Hyper-parameter tuning helps improve the performance of the satisfied group, but even made the performance of unsatisfied group worse.
- The performance imbalance between groups is more serious compared to the DT model

DECISION TREE VS. LOGISTIC REGRESSION

There is no significant difference of the overall performance of the two models, but decision tree did better than logistic regression in keeping a balance between two groups.

The possible reason might be that logistic regression assumes a linear relationship between features while decision tree does not. And from previous heatmap we could tell that our features do not have linear relationships.

Feature Importance

	feature	importance
0	income	0.37
1	purchase_amount	0.36
2	product_category_Food	0.04
3	region_West	0.03
4	product_category_Clothing	0.02
5	promotion_usage_0	0.02
6	loyalty_status_Gold	0.02
7	education_Bachelor	0.01
8	loyalty_status_Silver	0.01
9	product_category_Health	0.01
10	gender_Female	0.01

	feature	importance
0	income	0.38
1	purchase_amount	0.38
2	loyalty_status_Silver	0.01
3	purchase_frequency_occasional	0.01
4	purchase_frequency_frequent	0.01
5	promotion_usage_1	0.01
6	promotion_usage_0	0.01
7	product_category_Home	0.01
8	product_category_Health	0.01
9	product_category_Food	0.01
10	product_category_Electronics	0.01

- Both models give similar results about feature importance.
- Both top 2 features are the same, and both are numerical data: possible reason may be the explanatory power of categorical data has been weakened since they've been encoded.



Error Analysis

```
confusion_matrix(df_y2_test, prediction6)
```

```
array([[7734, 1439],
       [7658, 1542]])
```

```
confusion_matrix(df_y2_test, prediction7)
```

```
array([[7038, 2135],
       [6991, 2209]])
```

Both models have more false negatives than false positives, which indicates its lack of capability in predicting negative cases and distinguishing two groups.

After pulling the wrongly-predicted cases, we can further confirm that the model is inclined to predict unsatisfied cases as satisfied but not satisfied cases as unsatisfied.

Wrongly-predicted cases

income	purchase_amount	gender_Female	gender_Male	education_Bachelor	education_College	education_HighSchool	education_Masters	region_East	region_North	...	product_category_Food	product_category_Health	product_category_Home	promotion_usage_0	promotion_usage_1
0.858800	0.750893	1.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	...	0.0	1.0	0.0	1.0	0.0
0.787511	0.638010	0.0	1.0	0.0	1.0	0.0	0.0	1.0	0.0	...	0.0	0.0	0.0	1.0	0.0
0.179756	0.180190	0.0	1.0	0.0	1.0	0.0	0.0	1.0	0.0	...	0.0	0.0	0.0	1.0	0.0
0.506133	0.355400	1.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	1.0	0.0
0.777156	0.608833	1.0	0.0	0.0	1.0	0.0	0.0	0.0	1.0	...	1.0	0.0	0.0	0.0	1.0
0.778844	0.671311	0.0	1.0	0.0	0.0	0.0	1.0	1.0	0.0	...	0.0	0.0	0.0	0.0	1.0
0.944178	0.756829	1.0	0.0	0.0	0.0	1.0	0.0	0.0	1.0	...	0.0	0.0	0.0	1.0	0.0
0.269800	0.248151	0.0	1.0	1.0	0.0	0.0	0.0	1.0	0.0	...	0.0	0.0	0.0	0.0	1.0
0.508867	0.336285	0.0	1.0	0.0	0.0	0.0	1.0	0.0	0.0	...	0.0	1.0	0.0	1.0	0.0
0.764956	0.576488	1.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	1.0
0.542067	0.360028	1.0	0.0	0.0	1.0	0.0	0.0	1.0	0.0	...	1.0	0.0	0.0	0.0	1.0
0.152444	0.190653	0.0	1.0	1.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	1.0	1.0	0.0
0.792356	0.651240	1.0	0.0	0.0	1.0	0.0	0.0	0.0	1.0	...	1.0	0.0	0.0	0.0	1.0
0.875156	0.725741	0.0	1.0	0.0	1.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	1.0	0.0
0.358178	0.251069	1.0	0.0	0.0	1.0	0.0	0.0	1.0	0.0	...	1.0	0.0	0.0	1.0	0.0

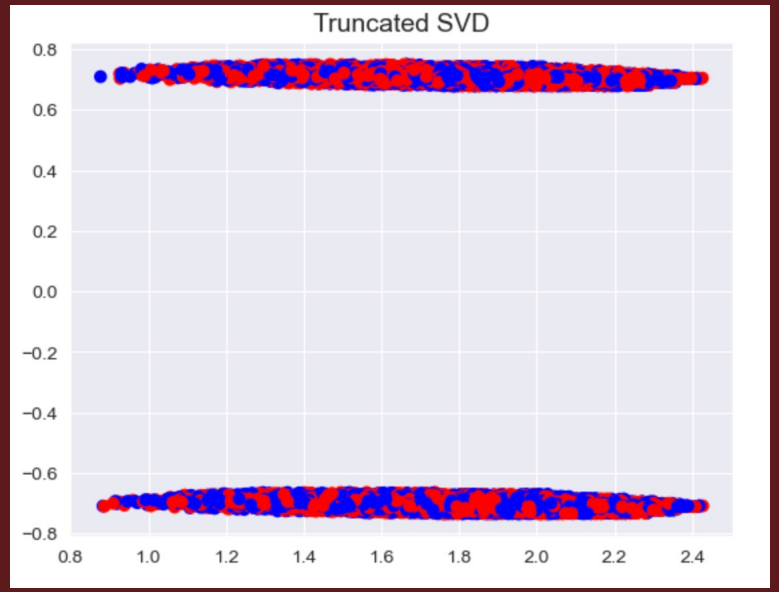
However, it's hard to find the pattern behind by observation(maybe there is no pattern). One possible reason might be that the 5 is not a suitable threshold for satisfaction and unsatisfaction, therefore the target variable and its prediction does not align with the ground-truth data features.

DIMENSION REDUCTION

1. Reduce features to 2-dimensional
2. Observe if there's any underlying pattern in features that is related to satisfaction.
3. Refit the DT and LR model with reduced features.

Xs labeled "satisfaction": blue

Xs labeled "unsatisfaction" : red



Unfortunately, there seems to be no significant information gain after plotting the dimension-reduced features.

And the refitting models still have a AUC-ROC score around 0.5

BAGGING

	precision	recall	f1-score	support
satisfied	0.50	0.98	0.66	9173
unsatisfied	0.49	0.02	0.04	9200
accuracy			0.50	18373
macro avg	0.49	0.50	0.35	18373
weighted avg	0.49	0.50	0.35	18373

- AUC-ROC=0.499
- The model performance of bagging is a little bit worse than the original model. The pros of bagging is that it helps in reducing variance by averaging the predictions of multiple models trained on different subsets of the data, which often leads to more stable and reliable predictions. Also bagging models are less prone to overfitting compared to single models, especially in high-variance models like decision trees. However, if the base learners are already biased towards a particular outcome, bagging might not significantly improve the model's performance. Our model might be of this kind of cases.

BOOSTING

	precision	recall	f1-score	support
satisfied	0.50	0.94	0.65	9173
unsatisfied	0.53	0.07	0.12	9200
accuracy			0.50	18373
macro avg	0.51	0.50	0.39	18373
weighted avg	0.51	0.50	0.39	18373

- AUC-ROC=0.502
- Boosting methods like AdaBoost have strength in better separation between classes and ultimately improve the AUC-ROC, since it could assign higher weights to misclassified instances in each iteration, which allows the algorithm to focus more on difficult-to-classify instances. However, boosting algorithms can be sensitive to noisy data and outliers as they tend to fit the data closely, which may lead to overfitting. Although its disadvantage is not that obvious in this case.



DISCUSSION

- The model performance isn’t that satisfying and this project may not answer our research question.
- The possible reason could be:
  - The data is simulated and it could not represent the real case.
  - The features may not fully represent customer features and product features.

